

BCI-MCP: Streaming Live EEG Band-Power Metrics to AI Assistants over the Model Context Protocol

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Abstract

Large-language-model (LLM) assistants increasingly call external tools through the Model Context Protocol (MCP). We describe BCI-MCP, an open-source (MIT) Python package that exposes live electroencephalography (EEG) from consumer and research devices to any MCP client. Its central design choice is that the model never sees raw EEG: a deterministic, documented pipeline (60 Hz notch, 1–45 Hz zero-phase band-pass, Welch power spectral density) reduces each 2 s window to band powers and six heuristic band-power ratios from the literature, together with artifact flags, a confidence score, and machine-readable statements of the pipeline’s limitations. We evaluate the released software on four questions. On synthetic signals with a known generator parameter, all six metrics track the parameter monotonically (Spearman $|\rho| = 0.996$). On real EEG from 109 subjects of the PhysioNet EEG Motor Movement/Imagery Dataset, the unmodified pipeline recovers the eyes-closed alpha increase (Berger effect): the *calm* metric rises in 105 of 109 subjects (Wilcoxon $p = 3 \times 10^{-19}$, rank-biserial $r = 0.99$; within-subject window AUC median 0.87). Injected artifacts are flagged at fixed amplitude thresholds, but the amplitude-only blink detector also fires on 40% of eyes-closed occipital windows, where genuine blinks are implausible; we report this limitation. A `get_brain_state` tool call over the stdio transport has a median round-trip latency of 1.3 ms. The metrics are heuristic proxies, not validated measures of cognitive state; we discuss what that implies for LLM-mediated interpretation of neural data.

1 Introduction

MCP is an open protocol through which an LLM application discovers and calls *tools*, reads *resources*, and uses *prompts* exposed by separate server processes [12]. It has made it simple to connect an assistant to a data source. Physiological data is an obvious candidate, and EEG in particular is now available from inexpensive headsets such as Muse and OpenBCI boards [10].

Connecting EEG to an LLM raises a specific risk. A language model asked to “read” a brain signal will produce a flu-

ent interpretation whether or not the data supports one. Handing a model raw samples invites it to invent spectral analysis; handing it a single “focus: 0.82” invites it to treat a heuristic ratio as a measurement. BCI-MCP is built around the opposite contract: all signal processing is deterministic and inspectable, the model only receives computed scalars, and every scalar arrives with the information needed to discount it (signal quality, artifact flags, a confidence value, a reliability status, the formula and literature basis of each metric, and a list of what the pipeline cannot detect).

This paper makes three contributions:

1. A device-agnostic, locally run MCP server for live EEG, packaged on PyPI and npm, with 14 tools, 2 resources and 1 prompt (Section 3).
2. An explicit “honesty layer”: limitations, formulas and caveats are served to the model as data, from a single source in the code.
3. A reproducible evaluation of the released pipeline on synthetic ground truth, on 109 subjects of public real EEG, on injected artifacts, and on end-to-end MCP latency, including negative results (Section 4). All scripts and outputs are in the repository.

2 Related work

General-purpose BCI and EEG software is mature. BCI2000 [17] and OpenViBE [16] are real-time BCI platforms; MNE-Python [7] is the standard for offline M/EEG analysis; BrainFlow [4] provides a uniform acquisition API across many biosensor boards; and the Lab Streaming Layer (LSL) [9] synchronizes streams between programs. BCI-MCP builds on BrainFlow and LSL for acquisition rather than replacing any of these tools. Its contribution is the interface to LLM assistants.

Connecting EEG to LLMs over MCP is not unique to this work. Neurosity announced a hosted MCP server for its Crown headset in September 2025 [14], before the current BCI-MCP server was written (the repository dates from March 2025, but the working MCP implementation dates from June 2026). Several other open-source projects appeared in 2026, including single-device servers for OpenBCI

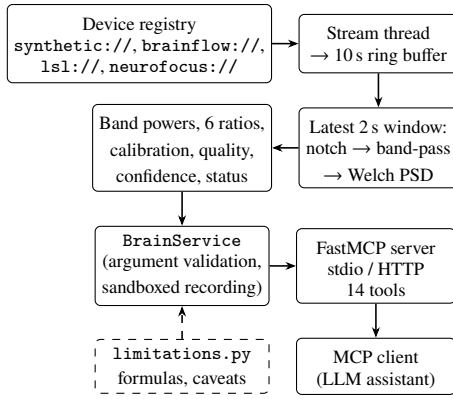


Figure 1: Data flow. The model receives only computed scalars, each with quality and confidence fields and the text of the pipeline’s limitations.

and Muse and an MCP wrapper for offline MNE analysis. BCI-MCP differs from these in three respects: it is device-agnostic behind one URI scheme, it runs locally with a synthetic device on the same code path as hardware, and it publishes an evaluation of its own pipeline. We do not claim to be first.

The metrics follow well-known EEG indices: the engagement index $\beta/(\alpha + \theta)$ of Pope et al. [15]; the theta/beta ratio used in attention research [11, 13], whose diagnostic value is contested [1]; the $(\theta + \alpha)/\beta$ drowsiness index of Eoh et al. [5]; and posterior alpha as the marker of relaxed wakefulness [2, 3].

3 System design

Acquisition. Every backend implements a five-method Device interface and registers a URI scheme. Supported sources are a synthetic generator, BrainFlow boards (OpenBCI, Muse and others), LSL streams, a generic serial device, the NeuroFocus dry-electrode ear-EEG prototype over USB serial and Bluetooth LE, and playback of recorded files. A single daemon thread drains the device into a 10 s multi-channel ring buffer (Figure 1).

Signal processing. On request, the pipeline takes the latest 2 s window. It applies a 60 Hz IIR notch ($Q = 30$) *before* a 1–45 Hz fourth-order Butterworth band-pass, both zero-phase (`filtfilt`), then estimates the power spectral density with Welch’s method [20] (≈ 1 s segments, 50% overlap) using SciPy [19] and NumPy [8]. Band power is the integral of the PSD over δ (1–4 Hz), θ (4–8), α (8–13), β (13–30) and γ (30–45 Hz), averaged over channels, with rectangular integration when a narrow band covers a single frequency bin.

Metrics. Six ratios are computed from θ , α and β only (Table 1). Delta and gamma are reported but excluded from every metric because, on consumer hardware over short windows, they are dominated by drift and muscle activity. Uncalibrated,

Table 1: Metrics exposed to the model. All are heuristic proxies.

Metric	Formula	Basis
focus	$\beta/(\alpha + \theta)$	Pope et al. [15]
attention	β/θ	inverse TBR [1, 13]
fatigue	$(\theta + \alpha)/\beta$	Eoh et al. [5]
engagement	β/α	arousal ratio
calm	$\alpha/(\alpha + \beta)$	alpha relaxation [2]
meditation	$\alpha/(\alpha + \beta + \theta)$	relative alpha [3]

each ratio is mapped to $[0, 1]$ by a logistic centred at a neutral value (1.0 for unbounded ratios, 0.5 for the two bounded ones). A 20 s calibration replaces this with a per-user z -score baseline.

Quality and confidence. Each window is checked for flatline (variance $< 10^{-3} \mu V^2$), railing (mean peak-to-peak $> 2000 \mu V$), probable EMG ($> 400 \mu V$ peak-to-peak) and blinks ($> 150 \mu V$ deviation on the first channel). Flatline, railing or “poor” quality mark the reading unreliable. Confidence is the product of the quality score, a calibration factor (1.0 calibrated, 0.6 not) and window fill, and is capped at 0.1 when unreliable.

Honesty layer. One module holds the analysis parameters, a list of seven limitations (for example, Welch averaging makes the pipeline blind to event-related potentials and spindles), and a one-line disclaimer. Every `get_brain_state` reply carries the disclaimer; `get_metric_definitions` and `get_pipeline_limitations` return the full text; and the `interpret_brain_state` prompt instructs the model to state those limits.

MCP interface and security. The server uses the official MCP Python SDK over stdio (local) or streamable HTTP. It exposes 14 tools (device listing and connection, brain state, band powers, signal quality, definitions, limitations, calibration, event marking, summary, recording, neurofeedback), the resources `brain://state` and `brain://device`, and one prompt. Because tool arguments come from a model that may be steered by untrusted content, all arguments are validated and bounded, recordings are confined to a sandbox directory with path-traversal and symlink checks, device schemes that grant file or port access (`playback://`, `serial://`) are refused over MCP, HTTP transports support bearer-token authentication, and loopback binds enable DNS-rebinding protection.

Engineering. The package supports Python 3.10–3.14, has 149 hardware-free tests run in CI, and is distributed on PyPI and as an npx launcher on npm. Documentation is at <https://enkhbold470.github.io/bci-mcp/>.

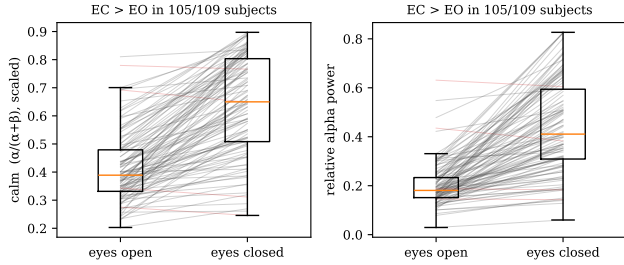


Figure 2: Per-subject means, eyes open vs. closed (109 subjects). Grey lines rise, red lines fall.

Table 2: Eyes closed (EC) vs. eyes open (EO), 109 subjects. Medians of per-subject means; Wilcoxon signed-rank; r = matched-pairs rank-biserial; AUC pools all windows.

Outcome	EO	EC	EC>EO	r	AUC
calm (primary)	0.39	0.65	105/109	+0.99	0.78
meditation	0.25	0.53	106/109	+0.99	0.80
relative alpha	0.18	0.41	105/109	+0.99	0.81
focus	0.43	0.36	15/109	-0.85	0.28

4 Evaluation

All experiments run the released pipeline code without modification. Scripts are in `paper/benchmarks/` and their JSON outputs in `paper/results/`; timings were measured on an Apple M5 Pro laptop (Python 3.12).

4.1 Implementation check on synthetic ground truth

The synthetic device mixes 10 Hz alpha, 20 Hz beta and 6 Hz theta sinusoids with Gaussian noise; its *focus* parameter trades alpha amplitude for beta amplitude. We swept the parameter from 0 to 1 in eleven steps with 50 windows each (un-calibrated). All six metrics were monotonic in the parameter in the expected direction (Spearman ρ = +0.996 for focus, attention and engagement; -0.996 for calm, fatigue and meditation). Scaled *focus* rose from 0.27 to 0.99 and *calm* fell from 0.92 to 0.10. This verifies the implementation, not the physiology.

4.2 Real EEG: the eyes-closed alpha increase

Closing the eyes increases posterior alpha power [2, 3], so a correct alpha-based pipeline must detect it. We used the PhysioNet EEG Motor Movement/Imagery Dataset, version 1.0.0 [6, 17, 18]: for each of 109 subjects, run 1 is a one-minute eyes-open baseline and run 2 a one-minute eyes-closed baseline (64 channels, 160 Hz). The method was fixed before looking at results: take O1, Oz and O2; split each run into non-overlapping 2 s windows (30 per run, 6,540 in total); run the unmodified, uncalibrated pipeline on each; and average per subject. The primary outcome was *calm*.

Calm was higher with eyes closed in 105 of 109 subjects (median difference +0.21, bootstrap 95% CI [0.17, 0.27]; Wilcoxon $W = 33$, $p = 3.2 \times 10^{-19}$), as were meditation and relative alpha (Table 2, Figure 2). *Focus*, whose denominator contains alpha, fell as expected. Within subjects, a single threshold on *calm* separated eyes-closed from eyes-open windows with a median AUC of 0.87 (IQR 0.72–0.97). Pooled across subjects without calibration, the AUC was 0.78, which illustrates why per-user calibration exists. No window was marked unreliable.

This is a check that the pipeline recovers a large, well-established effect, not evidence that *calm* measures calmness. The data are gel-electrode research recordings, not consumer headsets.

4.3 Artifact flagging

We injected artifacts of known amplitude into clean synthetic windows (4 channels at 256 Hz; 300 trials per cell). No clean window was flagged (0/300). Blinks (a 200 ms Gaussian on the first two channels) were flagged in 0% of trials at 100 μ V, 68% at 150 μ V and 100% from 200 μ V. Band-limited 30–100 Hz noise standing in for EMG was missed at 50 μ V RMS and flagged from 75 μ V RMS. Square-wave railing was marked unreliable from ± 1000 μ V, and constant-offset flat-lines in every trial.

Real data exposed a weakness. On the PhysioNet occipital windows, the blink detector fired on 22.6% of eyes-open and 39.5% of eyes-closed windows. People do not blink normally with their eyes closed, and occipital sites are far from the eyes, so these flags are almost certainly false positives, triggered by large alpha. The detector is a pure amplitude threshold on whichever channel is first, with no notion of montage or waveform. A blink flag lowers neither status nor confidence, so the metrics above are unaffected, but the flag itself mis-informs the model. Montage-aware, waveform-based blink detection is future work.

4.4 Latency

The per-window computation (filters, Welch, metrics, quality) had a median cost of 0.35 ms for 4 channels at 256 Hz, 0.42 ms for 8 channels at 250 Hz and 0.94 ms for 64 channels at 160 Hz (2,000 repetitions each). End to end, we launched `bci-mcp serve` as a subprocess and drove it with the official MCP Python client over stdio, as a desktop assistant would. Cold start to a completed `initialize` handshake took a median of 562 ms (10 runs). After `connect`, the first reading was available 387 ms later (the buffer needs 0.5 s of data). A `get_brain_state` call then had a median round trip of 1.30 ms (p95 2.34, p99 3.16 ms; 500 calls; median reply 1.2 kB). The pipeline therefore adds negligible latency compared with LLM inference. Readings describe the preceding 2 s, and acquisition adds a device-dependent delay (for example 125 ms per chunk for the synthetic device), which we did not measure on physical hardware.

5 Limitations and ethical considerations

The metrics are band-power ratios with literature precedent, not validated measures of focus, calm or fatigue, and the tool must not be used for medical or diagnostic purposes. The real-data evaluation covers one effect in one dataset of research-grade recordings. It does not establish performance on dry-electrode consumer devices, in motion, or for the attention and fatigue indices. Latency was measured with synthetic acquisition on one machine. Welch averaging makes the pipeline blind to transient events, and artifacts are flagged, not removed.

Neural data is sensitive. BCI-MCP runs locally by default and sends nothing anywhere unless the user connects it to a cloud-hosted assistant, at which point the computed metrics (not raw EEG) leave the machine with each tool call. Users should understand that before connecting. The larger risk is interpretive: an assistant may narrate a heuristic number as a fact about a person’s mind. The honesty layer reduces that risk, but whether models actually respect confidence and limitation fields is untested and is the most important open question for this kind of system.

6 Availability

Source (MIT): <https://github.com/enkhbold470/bci-mcp>. Install with `pip install bci-mcp` or run `npx -y bci-mcp`. The evaluation is reproduced by the scripts in `paper/benchmarks/`; the PhysioNet data is freely downloadable.

AI usage disclosure

Much of the June 2026 rewrite of BCI-MCP was written with AI coding assistants (Claude Code, Cursor) working from design documents kept in the repository; the author directed the design, integrated the NeuroFocus hardware protocol, reviewed and tested the changes, and is responsible for the code. The benchmark scripts and a first draft of this manuscript were also prepared with an AI assistant (Claude). The author checked every number against the benchmark outputs and every reference against its DOI record, and takes full responsibility for the content.

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